Priors and laws of motion for the two-series DSC model

One retained section: V, B_1, q_h, h_1, q_r, and r_1

For $n = 2$ observed series and dates $t = 1, 2, 3, 4$, this section records the complete prior hierarchy and the state-transition laws. All calibrated values are kept abstract. The subscripts on the hyperparameters identify which prior they belong to.

Priors and laws of motion

Let

$$\eta = \Big(\underbrace{\Psi, \nu_V}_{V \sim \mathrm{IW}_2(\Psi,\nu_V)} , \underbrace{m_B, C_B}_{B_1 \sim \mathcal{N}_2(m_B,C_B)} , \underbrace{a_h, b_h}_{q_{hj} \sim \mathrm{IG}(a_h,b_h)} , \underbrace{\mu_{h1}, \mu_{h2}, c_h}_{h_{1j}|q_{hj} \sim \mathcal{N}(\mu_{hj},(c_h+1)q_{hj})} , \underbrace{a_r, b_r}_{q_r \sim \mathrm{IG}(a_r,b_r)} , \underbrace{\mu_r, c_r}_{r_1|q_r \sim \mathcal{N}(\mu_r,(c_r+1)q_r)} \Big).$$

where $C_B \succ 0$, $\Psi \succ 0$, $\nu_V > 1$, $a_h, a_r, b_h, b_r > 0$, and $c_h, c_r > 0$. In the specification with one common IG shape,

$$a_h = a_r = a.$$

We condition on η throughout.

Mean states and their innovation covariance

$$\begin{aligned} V &\sim \mathrm{IW}_2(\Psi, \nu_V), & B_1 &\sim \mathcal{N}_2(m_B, C_B), \\ B_t \mid B_{t-1}, V &\sim \mathcal{N}_2(B_{t-1}, V), & t &= 2, 3, 4. \end{aligned}$$

Volatility states and their innovation variances

For $j = 1, 2$,

$$\begin{aligned} q_{hj} &\sim \mathrm{IG}(a_h, b_h), & h_{1j} \mid q_{hj} &\sim \mathcal{N}(\mu_{hj}, (c_h + 1)q_{hj}), \\ h_{tj} \mid h_{t-1,j}, q_{hj} &\sim \mathcal{N}(h_{t-1,j}, q_{hj}), & t &= 2, 3, 4. \end{aligned}$$

Correlation-coordinate state and its innovation variance

$$\begin{aligned} q_r &\sim \mathrm{IG}(a_r, b_r), & r_1 \mid q_r &\sim \mathcal{N}(\mu_r, (c_r + 1)q_r), \\ r_t \mid r_{t-1}, q_r &\sim \mathcal{N}(r_{t-1}, q_r), & t &= 2, 3, 4. \end{aligned}$$

Posterior after observing only y_1, y_2

Let

$$y_t \mid B_t, h_t, r_t \sim \mathcal{N}_2(B_t, \Sigma(h_t, r_t)), \qquad t = 1, 2,$$

where $h_t = (h_{t1}, h_{t2})'$ and r_t determines the 2×2 correlation. After observing y_1, y_2, the exact posterior kernel is

$$\begin{aligned} &p(B_{1:2}, h_{1:2}, r_{1:2}, V, q_{h1}, q_{h2}, q_r \mid y_{1:2}) \\ &\quad \propto \prod_{t=1}^{2} \mathcal{N}_2(y_t; B_t, \Sigma(h_t, r_t))\, \mathcal{N}_2(B_1; m_B, C_B) \mathcal{N}_2(B_2; B_1, V)\, \mathrm{IW}_2(V; \Psi, \nu_V) \\ &\qquad \times \prod_{j=1}^{2} \mathrm{IG}(q_{hj}; a_h, b_h) \mathcal{N}(h_{1j}; \mu_{hj}, (c_h + 1)q_{hj})\, \mathcal{N}(h_{2j}; h_{1j}, q_{hj}) \\ &\qquad \times \mathrm{IG}(q_r; a_r, b_r) \mathcal{N}(r_1; \mu_r, (c_r + 1)q_r)\, \mathcal{N}(r_2; r_1, q_r). \end{aligned}$$

Conditional on the mean states, the covariance posterior is conjugate:

$$V \mid B_{1:2}, y_{1:2} \sim \mathrm{IW}_2(\Psi + (B_2 - B_1)(B_2 - B_1)',\, \nu_V + 1)\,.$$

Conditional on the volatility path, the innovation-variance posteriors are

$$q_{hj} \mid h_{1:2,j}, y_{1:2} \sim \mathrm{IG}\left(a_h + 1,\, b_h + \frac{1}{2}\left[\frac{(h_{1j} - \mu_{hj})^2}{c_h + 1} + (h_{2j} - h_{1j})^2\right]\right), \qquad j = 1, 2,$$

and, conditional on the correlation path,

$$q_r \mid r_{1:2}, y_{1:2} \sim \mathrm{IG}\left(a_r + 1,\, b_r + \frac{1}{2}\left[\frac{(r_1 - \mu_r)^2}{c_r + 1} + (r_2 - r_1)^2\right]\right).$$

To use this posterior as the prior before observing y_3, propagate one step:

$$\begin{aligned} B_3 \mid B_2, V, y_{1:2} &\sim \mathcal{N}_2(B_2, V), \\ h_{3j} \mid h_{2j}, q_{hj}, y_{1:2} &\sim \mathcal{N}(h_{2j}, q_{hj}), \qquad j = 1, 2, \\ r_3 \mid r_2, q_r, y_{1:2} &\sim \mathcal{N}(r_2, q_r). \end{aligned}$$

This is the direct analogue of the original prior for (B_1, h_1, r_1) before any data are observed. Equivalently,

$$p(B_3, h_3, r_3, V, q_h, q_r \mid y_{1:2}) = \int p(B_3 \mid B_2, V) \prod_{j=1}^{2} p(h_{3j} \mid h_{2j}, q_{hj}) p(r_3 \mid r_2, q_r)$$

$$\times p(B_{1:2}, h_{1:2}, r_{1:2}, V, q_h, q_r \mid y_{1:2})\, dB_{1:2}\, dh_{1:2}\, dr_{1:2}.$$

Then the next update multiplies this predictive prior by

$$p(y_3 \mid B_3, h_3, r_3) = \mathcal{N}_2(y_3; B_3, \Sigma(h_3, r_3))\,.$$

Algorithm A: sampling given y_1, y_2

To draw from $p(B_{1:2}, h_{1:2}, r_{1:2}, V, q_{h1}, q_{h2}, q_r \mid y_{1:2})$, iterate the following Gibbs sampler with elliptical slice sampling (ESS) for the nonlinear volatility and correlation states. Given current $h_{1:2}$ and $r_{1:2}$, define $\Sigma_t = \Sigma(h_t, r_t)$ for $t = 1, 2$.

1. Draw the mean-state path $B_{1:2}$. Let $B = (B_1', B_2')'$ and draw

$$B \mid V, h_{1:2}, r_{1:2}, y_{1:2} \sim \mathcal{N}_4(M_B, \Omega_B),$$

where $\Omega_B = \Lambda_B^{-1}$,

$$\Lambda_B = \begin{pmatrix} C_B^{-1} + V^{-1} + \Sigma_1^{-1} & -V^{-1} \\ -V^{-1} & V^{-1} + \Sigma_2^{-1} \end{pmatrix},$$

and

$$M_B = \Omega_B \begin{pmatrix} C_B^{-1} m_B + \Sigma_1^{-1} y_1 \\ \Sigma_2^{-1} y_2 \end{pmatrix}.$$

2. Draw the mean innovation covariance.

$$V \mid B_{1:2}, y_{1:2} \sim \mathrm{IW}_2(\Psi + (B_2 - B_1)(B_2 - B_1)',\, \nu_V + 1)\,.$$

3. Draw the volatility innovation variances. For $j = 1, 2$,

$$q_{hj} \mid h_{1:2,j}, y_{1:2} \sim \mathrm{IG}\left(a_h + 1,\, b_h + \frac{1}{2}\left[\frac{(h_{1j} - \mu_{hj})^2}{c_h + 1} + (h_{2j} - h_{1j})^2\right]\right).$$

4. Draw the correlation innovation variance.

$$q_r \mid r_{1:2}, y_{1:2} \sim \mathrm{IG}\left(a_r + 1,\, b_r + \frac{1}{2}\left[\frac{(r_1 - \mu_r)^2}{c_r + 1} + (r_2 - r_1)^2\right]\right).$$

5. Update the volatility paths. For each $j = 1, 2$, update $h_{1:2,j} = (h_{1j}, h_{2j})'$ with ESS. Use the Gaussian prior

$$h_{1:2,j} \mid q_{hj} \sim \mathcal{N}_2\left(\mu_{hj}\mathbf{1}_2,\, q_{hj}\begin{pmatrix} c_h+1 & c_h+1 \\ c_h+1 & c_h+2 \end{pmatrix}\right),$$

and the ESS likelihood factor

$$L_h(h_{1:2,j}) = \prod_{t=1}^{2} \mathcal{N}_2(y_t; B_t, \Sigma(h_t, r_t)),$$

holding the other volatility coordinate and $r_{1:2}$ fixed.

6. Update the correlation path. Update $r_{1:2} = (r_1, r_2)'$ with ESS. Use the Gaussian prior

$$r_{1:2} \mid q_r \sim \mathcal{N}_2\left(\mu_r\mathbf{1}_2,\, q_r\begin{pmatrix} c_r+1 & c_r+1 \\ c_r+1 & c_r+2 \end{pmatrix}\right),$$

and the ESS likelihood factor

$$L_r(r_{1:2}) = \prod_{t=1}^{2} \mathcal{N}_2(y_t; B_t, \Sigma(h_t, r_t)),$$

holding $h_{1:2}$ fixed.

After convergence, retain the joint draws of $(V, q_h, q_r, B_{1:2}, h_{1:2}, r_{1:2})$ as a Monte Carlo approximation to the posterior. Keep all components of each draw together for the following update.

Algorithm B: updating with y_3

Algorithm A supplies the saved draws conditional on $y_{1:2}$. Algorithm B has four steps: extend, weight, resample, and update the parameters and full paths using Gibbs and ESS.

Write $\theta = (V, q_{h1}, q_{h2}, q_r)$, $x_t = (B_t, h_t, r_t)$, and $\pi_t(\theta, x_{1:t}) = p(\theta, x_{1:t} \mid y_{1:t})$. Suppose we have retained M equally weighted joint draws

$$(\theta^{(m)}, x_{1:2}^{(m)}), \qquad m = 1, \dots, M, \qquad \text{from } \pi_2.$$

For this two-series model, the observation covariance is

$$\Sigma(h_t, r_t) = D_t \begin{pmatrix} 1 & \tanh r_t \\ \tanh r_t & 1 \end{pmatrix} D_t, \qquad D_t = \operatorname{diag}(e^{h_{t1}/2}, e^{h_{t2}/2}).$$

The exact Bayesian update is

$$\boxed{\pi_3(\theta, x_{1:3}) = \frac{\mathcal{N}_2(y_3; B_3, \Sigma(h_3, r_3))\, p(x_3 \mid x_2, \theta)\, \pi_2(\theta, x_{1:2})}{p(y_3 \mid y_{1:2})}.}$$

Substituting $\pi_2 \propto p(\theta, x_{1:2}) \prod_{t=1}^{2} p(y_t \mid x_t)$ gives $\pi_3 \propto p(\theta, x_{1:3}) \prod_{t=1}^{3} p(y_t \mid x_t)$, the same posterior obtained by using the original priors and all three observations.

1. Extend each saved draw to date 3. For $m = 1, \dots, M$, keep $\theta^{(m)}$ and $x_{1:2}^{(m)}$ together and draw independently conditional on that saved draw:

$$\begin{aligned} B_3^{(m)} &\sim \mathcal{N}_2(B_2^{(m)}, V^{(m)}), \\ h_{3j}^{(m)} &\sim \mathcal{N}(h_{2j}^{(m)}, q_{hj}^{(m)}), \qquad j = 1, 2, \\ r_3^{(m)} &\sim \mathcal{N}(r_2^{(m)}, q_r^{(m)}). \end{aligned}$$

The resulting complete draw is

$$\left(V^{(m)}, q_{h1}^{(m)}, q_{h2}^{(m)}, q_r^{(m)}, B_{1:3}^{(m)}, h_{1:3}^{(m)}, r_{1:3}^{(m)}\right),$$

where m labels the complete draw, including all parameters and paths. Superscripts label Monte Carlo draws, while subscripts $1:3$ label time.

2. Weight using only the new observation. Compute

$$\ell_m = \log \mathcal{N}_2\left(y_3; B_3^{(m)}, \Sigma(h_3^{(m)}, r_3^{(m)})\right), \qquad w_m = \frac{\exp(\ell_m - \ell_{\max})}{\sum_{k=1}^{M} \exp(\ell_k - \ell_{\max})},$$

where $\ell_{\max} = \max_k \ell_k$. The proposal in step 1 is $\pi_2(\theta, x_{1:2}) p(x_3 \mid x_2, \theta)$. The target-to-proposal ratio leaves precisely the likelihood of y_3 as the unnormalized importance weight. Each complete draw has weight w_m, and together the weighted complete draws approximate the updated posterior.

3. Resample complete draws. For each new draw $i = 1, \dots, M$, choose an index $A_i \in \{1, \dots, M\}$ with $\Pr(A_i = m) = w_m$. Copy into new draw i the complete tuple

$$\left(V^{(A_i)}, q_{h1}^{(A_i)}, q_{h2}^{(A_i)}, q_r^{(A_i)}, B_{1:3}^{(A_i)}, h_{1:3}^{(A_i)}, r_{1:3}^{(A_i)}\right).$$

For example, $A_i = 7$ means copy all parameters and paths from draw 7 together. Selection is with replacement, so duplicate new draws are allowed. The observation y_3 therefore changes the distribution of the old states and parameters through their weights.

Continue to Step 4 on page 5.

Monte Carlo accuracy. The boxed identity is exact. A finite saved sample, weighted or resampled, is an approximation to it. Step 4 moves the parameters and all three dates continuously, with π_3 as the invariant distribution. A finite number of moves does not guarantee exact independent posterior draws; convergence and mixing still need to be checked.

Algorithm B (continued): Step 4

Update the parameters and full paths using Gibbs and ESS.

Use each resampled complete draw as starting values, perform the following full Gibbs/ESS sweep R times, and retain its final parameters and paths. In every block, use the most recently updated values of the other variables. The original hyperparameters η remain fixed.

(a) Draw $B_{1:3}$ jointly. Set $\Sigma_t = \Sigma(h_t, r_t)$, $B = (B_1', B_2', B_3')'$, and draw

$$B \mid V, h_{1:3}, r_{1:3}, y_{1:3} \sim \mathcal{N}_6(M_{B,3}, \Omega_{B,3}), \qquad \Omega_{B,3} = \Lambda_{B,3}^{-1},$$

$$\Lambda_{B,3} = \begin{pmatrix} C_B^{-1} + V^{-1} + \Sigma_1^{-1} & -V^{-1} & 0 \\ -V^{-1} & 2V^{-1} + \Sigma_2^{-1} & -V^{-1} \\ 0 & -V^{-1} & V^{-1} + \Sigma_3^{-1} \end{pmatrix}, \qquad M_{B,3} = \Omega_{B,3} \begin{pmatrix} C_B^{-1} m_B + \Sigma_1^{-1} y_1 \\ \Sigma_2^{-1} y_2 \\ \Sigma_3^{-1} y_3 \end{pmatrix}.$$

(b) Draw V. With $\Delta B_t = B_t - B_{t-1}$,

$$V \mid B_{1:3}, y_{1:3} \sim \mathrm{IW}_2\left(\Psi + \sum_{t=2}^{3} \Delta B_t \Delta B_t',\, \nu_V + 2\right).$$

(c) Draw q_{h1}, q_{h2}, q_r. For $j = 1, 2$,

$$q_{hj} \mid h_{1:3,j}, y_{1:3} \sim \mathrm{IG}\left(a_h + \frac{3}{2},\, b_h + \frac{1}{2}\left[\frac{(h_{1j} - \mu_{hj})^2}{c_h + 1} + \sum_{t=2}^{3} (h_{tj} - h_{t-1,j})^2\right]\right),$$

$$q_r \mid r_{1:3}, y_{1:3} \sim \mathrm{IG}\left(a_r + \frac{3}{2},\, b_r + \frac{1}{2}\left[\frac{(r_1 - \mu_r)^2}{c_r + 1} + \sum_{t=2}^{3} (r_t - r_{t-1})^2\right]\right).$$

(d) Update the full volatility and correlation paths using ESS. Define

$$K_3(c) = \begin{pmatrix} c+1 & c+1 & c+1 \\ c+1 & c+2 & c+2 \\ c+1 & c+2 & c+3 \end{pmatrix}.$$

For each $j = 1, 2$, perform an ESS update of $h_{1:3,j}$ with Gaussian prior $\mathcal{N}_3(\mu_{hj}\mathbf{1}_3, q_{hj}K_3(c_h))$, holding the other volatility coordinate and $r_{1:3}$ fixed. Then perform an ESS update of $r_{1:3}$ with Gaussian prior $\mathcal{N}_3(\mu_r\mathbf{1}_3, q_rK_3(c_r))$, holding $h_{1:3}$ fixed. For each update, the likelihood factor is

$$L_{1:3}(B, h, r) = \prod_{t=1}^{3} \mathcal{N}_2(y_t; B_t, \Sigma(h_t, r_t)).$$

These updates change the states at dates 1 and 2 as well as date 3, and therefore smooth the whole path conditional on $y_{1:3}$.

Which data are needed? Steps 1–3 need only the saved joint draws and y_3. Step 4 also needs y_1, y_2 to evaluate the posterior at new paths. This does not count the old data twice: each Gibbs or ESS move targets $p(\theta, x_{1:3})L_{1:3}$, in which each observation appears once.

This is a resample–move update (Gilks and Berzuini, 2001), using elliptical slice sampling for the Gaussian-prior path blocks (Murray, Adams and MacKay, 2010).